- Persamaan Garis

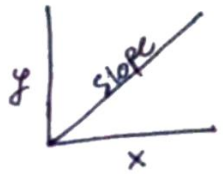
(Materi ini fokusnya ke graph bawah)

$$\boxed{y = mx + c}$$

→ y = nilai $mx + c$

→ m (gradien) = Rasio besaran y bila x berubah

→ c (intercept) = Nilai y saat x bernilai 0



(A) Penalaran

→ $x = 0$ & $y = 1$
$x = 1$ & $y = 3$ } Berubah (2) satuan maka $m = 2$

(B) Jika Melalui 2 titik

→ Jika lewat x_1, y_1 & x_2, y_2 maka :

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

- Persamaan Kuadrat

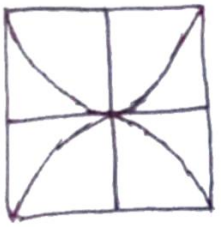
→ Definisi Formal

$$\boxed{y = ax^2 + bx + c}$$

a, b, c → Koefisien

→ $a > 0$
↳ Grafik keatas & minimum

→ $a < 0$
↳ Grafik bekas bawah & maksimum



→ $a > 0$

→ $a < 0$

- Akar Persamaan Kuadrat

→ Bentuknya $ax^2 + bx + c = 0$ maka x adalah akar Per. Kuadrat

→ Cara dapetin x Pake Diskriminan

$$x = \frac{-b \pm \sqrt{D}}{2a} \quad (D = b^2 - 4ac)$$

$D > 0$ → Dua akar berbeda (Hitung ± x dapatin x_1, x_2)
$D = 0$ → Satu akar (akar kembar)
$D < 0$ → Dua akar Kompleks

(Sama aja)

→ cari maksimum / minimum

$$x = \frac{-b}{2a}$$ (rumusnya sama)

→ Kalo $a > 0$ maka minimum

→ Kalo $a < 0$ maka maksimum

- Persamaan Lingkaran

  2 kondisi :

  → titik Pusat (a,b) & jari-jari r

  ↳ Persamaan $(x-a)^2 + (y-b)^2 = r^2$

  a, b → <u>koordinat</u>

  → Pusat $(0,0)$

  ↳ Persamaan : $x^2 + y^2 = r^2$

- Akar Kuadrat

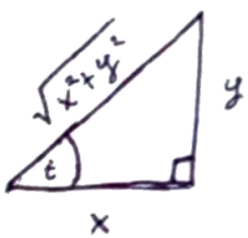
  → $\sqrt{y} = x \quad (x^2 = y)$

  → Non-negatif x

  - $\sqrt{x}$ → Kuadrat utama
  - $-\sqrt{x}$ → Negatif

  → $\sqrt{x} = x^{\frac{1}{2}}$

- Sinus Cosinus



Sin → y Axis

Cos → x Axis

$$\sin(t) = \frac{y}{\sqrt{x^2+y^2}}$$

$$\cos(t) = \frac{x}{\sqrt{x^2+y^2}}$$

bersan bisa konversi ke real

$\pi = 180^\circ$ dan seterusnya

(Cara ngitungnya pake misal <u>1</u>)

Kaya di <u>Khan</u> <u>Akademy</u>

PS : Udah dipelajari di <u>Trigonometry</u> <u>Khan Akademy</u> tapi lupa hehe